M. Vignesh

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Chennai, Tamil Nadu, India linkedin github Portfolio

Summary

AI/ML engineering student specializing in building real-time intelligent systems and applied AI solutions. Experienced in developing end-to-end machine learning pipelines, including simulation-driven digital twin architectures and multi-modal AI assistants. Demonstrated ability to design scalable systems, optimize model performance, and translate AI concepts into deployable, real-world applications across automation and intelligent interfaces.

Education

RMK Engineering College
B.Tech AI & DS (2023–2027)
CGPA: 7.80

Sri Ram Dayal Khemka Vivekananda Vidyalaya
Class XII – 80.2%
Class X – 84.2%

Achievements

KPIT Sparkle 2026 Finalist
Top 16 / 18,114+ teams
Led the development of a Digital Twin-based motion planning system integrating simulation-driven optimization and real-time decision-making for dynamic environments.

CMRIT 24hr Hackathon
Designed and developed an AI-driven eco-driving and emission estimation system under real-time constraints, focusing on efficiency and sustainability.

BIT Paper Presentation
Presented technical insights on AI-driven intelligent systems and emerging architectures, demonstrating applied understanding of modern AI paradigms.

Skills

Languages: Python, C++, Java
Frameworks: TensorFlow, FastAPI
AI/ML: NLP, Deep Learning, Computer Vision, XAI
Data Tools: Pandas, NumPy
Tools: Git, GitHub, Colab, API Integration

Certifications

NPTEL, Oracle AI Foundations, Qualcomm AI, Infosys ML

Projects

Digital Twin Motion Planner
Designed and implemented a simulation-driven autonomous navigation system for dynamic environments, integrating trajectory optimization, sensor fusion, and real-time decision-making. The system focuses on sim-to-real validation, enabling adaptive motion planning with improved efficiency and robustness in changing conditions.

ORION – Multi-Modal AI Assistant
Developed a scalable multi-modal AI assistant leveraging large language model orchestration with a FastAPI backend for real-time task execution. Incorporated contextual memory and modular architecture, enabling efficient handling of multi-input workflows and improved response relevance.

UNORA_MUSIC – AI Media Platform
Engineered a Flutter-based intelligent media platform integrating a local Mistral model via Ollama to enable natural language-driven interaction. Implemented real-time audio processing and personalized recommendation features, enhancing user experience through AI-powered control systems.

Eco-Driving and Pollution Tracker
Developed machine learning models to analyze driving behavior and estimate emissions, enabling real-time evaluation of driving efficiency and environmental impact. The system provides data-driven insights for sustainable mobility optimization.

Skin Disease Diagnosis (XAI)
Built a deep learning-based diagnostic system using MobileNetV2 for image classification, enhanced with Grad-CAM visualization to provide interpretable predictions. Focused on improving model transparency and reliability in medical decision support scenarios.

Experience

Infosys Springboard Internship Aug 2025 – Oct 2025
Developed a machine learning model for shipment time prediction by performing data preprocessing, feature engineering, and model optimization. Improved prediction accuracy by 15%, contributing to enhanced logistics planning and operational efficiency.

AICTE Shell Edunet Internship Jun 2025 – Jul 2025
Worked on AI and data analytics pipelines focused on sustainability use cases, involving data modeling, analytical processing, and insight generation to support informed decision-making.